

Question ID: 1c29bfd1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The pressure exerted on a scuba diver at sea level is **14.70 pounds per square inch (psi)**. For each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by **0.44 psi**. What is the total pressure, in **psi**, exerted on the scuba diver at **105** feet below sea level?

Answer

- A. **60.90**
- B. **31.50**
- C. **14.70**
- D. **0.44**

Correct Answer: A

Rationale

Choice A is correct. It's given that the pressure exerted on a scuba diver at sea level is **14.70 pounds per square inch (psi)**. It's also given that for each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by **0.44 psi**. The total pressure, in **psi**, exerted on the scuba diver at ***x*** feet below sea level can be represented by the expression **$0.44x + 14.70$** . Substituting **105** for ***x*** in this expression yields **$0.44(105) + 14.70$** , or **60.90**. Therefore, the total pressure exerted on the scuba diver at **105** feet below sea level is **60.90 psi**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pressure, in **psi**, exerted on the scuba diver at sea level, not at **105** feet below sea level.

Choice D is incorrect. This is the rate by which the pressure, in **psi**, exerted on the scuba diver increases for each foot the scuba diver descends below sea level.